

Lecture 1: Introduction

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Outline

Introduction

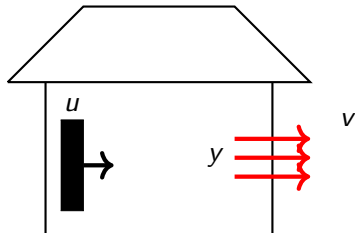
The transfer function, poles & zeros, stability

Block diagram representation

What's next

Today's Ordlista

English	Svenska
Laplace transform	Laplacetransform
denominator	nämnarpolynomet
numerator	täljarpolynomet
convolution	faltning
final value (theorem)	slutvärdessatsen
transfer function	överföringsfunktion
characteristic equation	karakteristisk ekvation
roots	rötter
left half plane (LHP)	vänster halvplan (VHP)
right half plane (RHP)	höger halvplan (HHP)
pole	pol
zero	nollställe
impulse response	impulssvar
feedback	återkoppling
disturbance	störning
loop gain	kretsförstärkning, öppna systemet
open-loop system	öppet system, kretsförstärkning

$$\begin{cases} y &= \text{indoor temperature} \\ u &= \text{power supplied to the system} \\ v &= \text{outside temperature} \\ r &= \text{desired temperature} \end{cases}$$


- Consider the heat balance equation: $C \frac{\Delta T}{\Delta t} = \text{power supplied} - \text{power demand}$
- In our notation, Newton's law of cooling gives: $C \frac{d}{dt} y(t) = u(t) - D(y(t) - v(t))$

$$\dot{y}(t) + \alpha y(t) = \beta u(t) + \alpha v(t)$$

The course addresses the control (regulation) of

Dynamical Systems = Differential equations!

These are dealt with (in the first part of the course) with the help of

Laplace Transform

$$Y(s) = \mathcal{L}\{y(t)\} = \int_0^{\infty} y(t)e^{-st} dt$$

Properties:

- i. $\mathcal{L}\left\{\frac{d}{dt} y(t)\right\} = sY(s) - y(0)$
- ii. $\mathcal{L}\left\{\int_0^t y(\tau) d\tau\right\} = \frac{1}{s} Y(s)$
- iii. Convolution: $\mathcal{L}\left\{\int_0^t g(\tau)u(t-\tau) d\tau\right\} = G(s)U(s)$
- iv. Final value theorem: $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$, if $\lim_{t \rightarrow \infty} y(t)$ exists [Resultat 2.1, p. 35]



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The transfer function, poles & zeros, stability

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Connection to differential equations: The transfer function

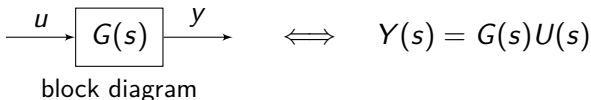
Example

$$\begin{cases} \ddot{y} + a_1\dot{y} + a_2y = b_0\dot{u} + b_1u \\ y(0) = \dot{y}(0) = u(0) = 0 \end{cases}$$

$$\xrightarrow{\mathcal{L}} s^2 Y(s) + a_1 s Y(s) + a_2 Y(s) = b_0 s U(s) + b_1 U(s)$$

$$Y(s) = \underbrace{\frac{b_0 s + b_1}{s^2 + a_1 s + a_2}}_{\text{transfer function, } G(s)} U(s)$$

The transfer function describes the relationship between the input and the output



Observe that: Initial values = 0

Definition

The system's *poles* are given by the roots of the denominator of the transfer function.

Example

$$s^2 + a_1 s + a_2 = 0$$

Application: (simplified) Inverted pendulum [Fig. 3.6, Åström och Murray]

$$s^2 + g/l = 0 \quad \Rightarrow \quad \text{Has poles in } s = \pm \sqrt{g/l}$$

Poles: LHP and RHP

Example

If we let $u = 0$ in our linear diff. eq., the solution of the corresponding homogeneous system is $y_0(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$, $\lambda_1 \neq \lambda_2$, where λ_1, λ_2 are the *roots* of the *characteristic equation*: $s^2 + a_1 s + a_2^*$.

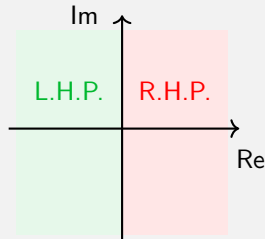
Note that complex roots can also come in complex conjugate pairs:

$\lambda = x + i\omega$ and $\bar{\lambda} = x - i\omega$, where x is the real part and ω the imaginary part.

We can write the exp term as:

$$e^{\lambda t} = e^{xt} [\cos(\omega t) + i \sin(\omega t)]$$

$$\xrightarrow{t \rightarrow \infty} \begin{cases} 0, & \text{Re}[\lambda] < 0, \text{ Left Half Plane, } \textit{Stable} \\ \infty, & \text{Re}[\lambda] > 0, \text{ Right Half Plane, } \textit{Unstable} \end{cases}$$



*roots = system's poles (see example in slide 7)

Poles: Exercise

Example

Task: Determine the output signal of the system

$$\dot{y} + y = 20, y(0) = 0$$

Solution:

$$\begin{aligned} &\xrightarrow{\mathcal{L}} sY(s) - y(0) + Y(s) = \frac{20}{s} \\ \Rightarrow Y(s) &= \underbrace{\frac{20}{s}}_{U(s)} \underbrace{\frac{1}{s+1}}_{G(s)} = 20 \left(\frac{1}{s} - \frac{1}{s+1} \right) \\ &\xrightarrow{\mathcal{L}^{-1}} y(t) = 20(1 - e^{-t}) \xrightarrow{t \rightarrow \infty} 20 \end{aligned}$$

Pole in -1 : LHP, therefore stable

Note: Here $u(t) = 20$ for all $t \geq 0$, i.e., u is a **step** of amplitude 20. **Homework:** Derive $U(s)$

Homework: Use the final value theorem to derive $y(t) \xrightarrow{t \rightarrow \infty} 20$



The system's *zeros* are the roots of the numerator of the transfer function.

$$b_0s + b_1 = 0 \implies \text{zero in } s = -b_1/b_0$$

Intuition: Zeros in the RHP $\Rightarrow \frac{1}{G(s)}$ is unstable $\Rightarrow$ Difficult control problem.



Convolution: $\xRightarrow{\mathcal{L}^{-1}} y(t) = \int_0^t g(\tau)u(t - \tau) \, d\tau$

The function $g(t)$ is called the **impulse response** (if $u = \delta(t)$ then $y(t) = g(t)$).

If $g(\tau) = 1$ then $y = \int u$. Hence, $G(s) = \frac{1}{s} =$ integrator

Input-Output (I/O) stability

Theorem ([Sats 2.1])

A system is *Input-Output (I/O) stable* if and only if $\int_0^\infty |g(\tau)| d\tau < \infty$.

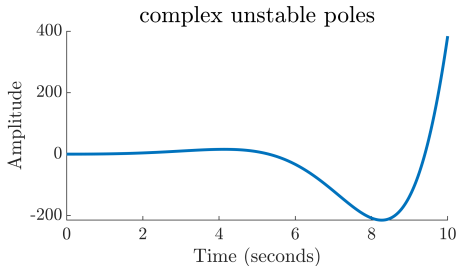
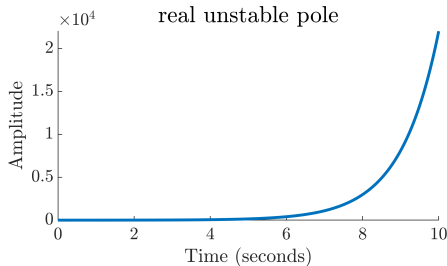
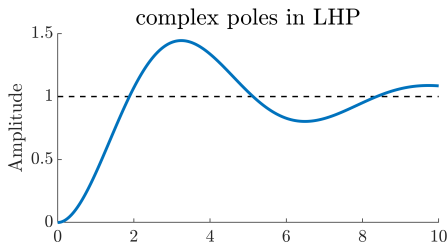
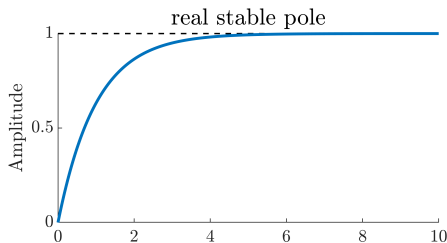
Theorem ([Sats 2.2])

A system with a proper transfer function $G(s)$ is *Input-Output (I/O) stable* if and only if all the poles of $G(s)$ are strictly in the LHP.

* in the course, we refer to “Input-Output (I/O) stable” as “stable”

Examples of 1st and 2nd order systems

The plots depict the **step response** of a system, i.e., $y(t)$ when $u(t) = 1$ unit step





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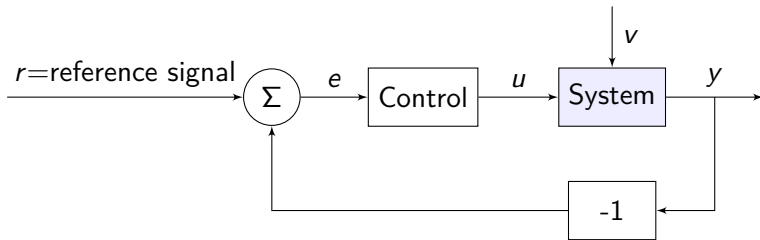
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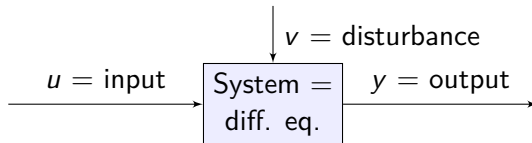
Block diagram representation

What's next

Block diagram: Feedback

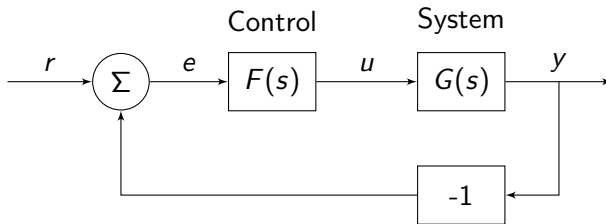


The *control error* is given by $e = r - y$.



Block diagram: Equations

Assume for the moment that no disturbance acts on the system



$$\Rightarrow \begin{cases} Y(s) = G(s)U(s) = G(s)F(s)E(s) \\ E(s) = R(s) - Y(s) \end{cases}$$

$$\Rightarrow Y = GFR - GFY$$

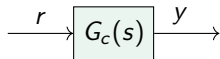
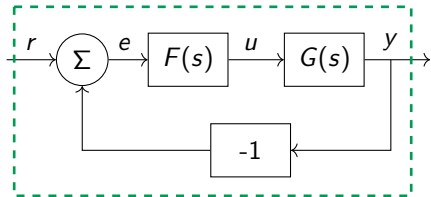
$$\Rightarrow (1 + GF)Y = GFR$$

From reference to output: The closed-loop transfer function

$$(1 + GF)Y = GFR$$

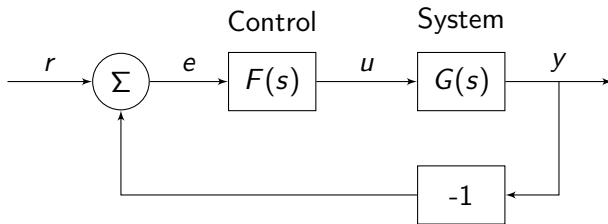
From this, we obtain the expression for how $r(t) \mapsto y(t)$.

$$\Rightarrow Y(s) = \underbrace{\frac{G(s)F(s)}{1 + G(s)F(s)}}_{G_c(s)} R(s)$$



$G_c(s)$ is the **closed-loop system**'s transfer function [p. 60-61]

From reference to control error and reference to input



► **From reference to control error.**

From $E(s) = R(s) - Y(s)$ and $Y(s) = G(s)F(s)E(s)$, we obtain the expression for how $r(t) \mapsto e(t)$.

$$\Rightarrow E(s) = \frac{1}{1 + G(s)F(s)} R(s)$$

► **From reference to input.**

From $U(s) = F(s)E(s)$, we obtain the expression for how $r(t) \mapsto u(t)$.

$$\Rightarrow U(s) = \frac{F(s)}{1 + G(s)F(s)} R(s)$$

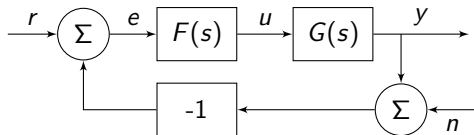
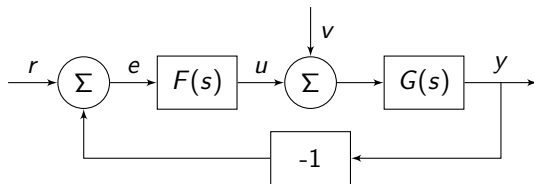
From input disturbance to output and output disturbance to output

- **From input disturbance to output, $v(t) \mapsto y(t)$:**

$$\frac{G(s)}{1 + G(s)F(s)}$$

- **From output disturbance to output, $n(t) \mapsto y(t)$:**

$$-\frac{G(s)F(s)}{1 + G(s)F(s)}$$



We will see this later in the course.



⇒ The poles of the closed-loop system are given by

This determines the stability of the closed-loop system!



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Next time: PID Control

Homework:

Repeat Laplace and differential equations!

Suggested exercises (not included in the Övningar!): 1.3, 1.5, 1.4, 1.7